DATA ANALYTICS WITH TABLEAU

Group Project Report

Visualizing Housing Market Trends:

An Analysis of Sale Prices and Features Using Tableau

Program	Data Analytics with Tableau
Platform	SkillWallet
Project Type	Group Project — Data Visualization & Web Integration
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PROJECT TEAM

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1. Introduction & Problem Statement

1.1 Background

The real estate market is highly dynamic, influenced by a multitude of factors including property size, location, renovation history, structural characteristics, and market timing. For companies like ABC Company, making data-driven decisions about pricing strategies, investment priorities, and customer targeting requires a clear and analytical understanding of the patterns underlying property sales data.

This project was undertaken as part of the Data Analytics with Tableau program on the SkillWallet platform. The team of five students was tasked with analyzing a comprehensive housing dataset and presenting insights through interactive Tableau visualizations, culminating in a web-integrated dashboard.

1.2 Problem Statement

ABC Company faces challenges in:

- Understanding which factors most significantly influence house sale prices
- Determining the impact of renovation history on current market value
- Assessing the distribution of housing inventory by age, bedroom/bathroom count, and floor configuration
- Making informed strategic decisions around pricing, marketing, and investment

Without clear visual analytics, these insights remain buried in raw data tables. The goal of this project is to surface them through well-designed Tableau dashboards and make them accessible through a Flask-based web application.

1.3 Objectives

- Analyze 21,609 housing transaction records to identify pricing trends
- Visualize the relationship between renovation recency and sale price
- Examine the age distribution of homes by renovation status
- Understand how house features (bathrooms, bedrooms, floors) vary by house age
- Build and publish an interactive Tableau dashboard on Tableau Public
- Embed the dashboard in a Flask web application for broad accessibility

1.4 Stakeholders

Stakeholder	Interest / Use Case
Real Estate Analysts	Deep-dive into pricing trends and renovation ROI
Marketing Teams	Identify target demographics and property features in demand
Company Executives	Strategic decisions on portfolio pricing and investments
Data Analytics Students	Learning project demonstrating full analytics pipeline

2. Dataset Overview

2.1 Source & Description

The dataset used in this project — Transformed_Housing_Data2.csv — contains 21,609 records of residential property sales. It has been pre-processed and feature-engineered to include derived attributes such as house age, years since renovation, and condition categories.

File Name: Transformed_Housing_Data2.csv
Total Records: 21,609
Total Columns: 31
Source: Real estate transaction records (transformed for analytics)
Format: CSV (Comma-Separated Values)

2.2 Key Variables

Column	Type	Description
Sale_Price	Numeric	Final sale price of the property ($)
No of Bedrooms	Numeric	Number of bedrooms in the house
No of Bathrooms	Numeric	Number of bathrooms (including half-baths)
Flat Area (in Sqft)	Numeric	Total living area in square feet
Lot Area (in Sqft)	Numeric	Total lot/land area in square feet
No of Floors	Numeric	Number of floors in the building
Overall Grade	Ordinal (1-13)	Construction and design quality grade
Age of House (in Years)	Numeric	Calculated age of the property
Years Since Renovation	Numeric	Years elapsed since last renovation (0 = never)
Ever_Renovated_Yes	Binary (0/1)	1 = house has been renovated at least once
Condition_of_the_House_*	Binary Encoded	One-hot encoded condition: Excellent, Fair, Good, Okay
Waterfront_View_Yes	Binary (0/1)	Property has a waterfront view
Zipcode_Group_*	Binary Encoded	One-hot encoded geographic zones (Groups 1–9)
Latitude / Longitude	Numeric	Geographic coordinates of the property

2.3 Summary Statistics

Metric	Value	Notes
Total Records	**21,609**	Unique property transactions
Average Sale Price	**$511,619**	Mean across all records
Minimum Sale Price	$75,000	Lowest recorded transaction
Maximum Sale Price	$1,129,575	Highest recorded transaction
Total Basement Area	38,643,798 sqft	Cumulative across all properties
Average Bedrooms	3.37	Per property
Average Bathrooms	2.11	Per property
Renovated Properties	914 (4.2%)	Ever_Renovated_Yes = 1
Non-Renovated Properties	20,695 (95.8%)	Majority of the dataset

3. Tools & Technologies Used

Tool / Technology	Purpose	Details
Tableau Desktop / Public	Data visualization & dashboard	Primary BI tool; published to Tableau Public
Python (Pandas)	Data preparation & EDA	CSV parsing, stat computation, data cleaning
Flask	Web application framework	Python-based backend; renders HTML templates
HTML / CSS / JavaScript	Frontend web interface	Responsive pages; Tableau JS embed API
Tableau JS Embed API	Dashboard embedding	Embeds live Tableau dashboard in the web app
Microsoft Excel / CSV	Data storage & exploration	Source data file: Transformed_Housing_Data2.csv
SkillWallet Platform	Project management	Kanban, epics, stories, mentoring
Git / GitHub	Version control	Source code management and collaboration

4. Data Collection

4.1 Dataset Acquisition

The housing dataset was sourced as part of the SkillWallet project brief for ABC Company. It represents a transformed version of a publicly available real estate dataset, enhanced with derived features specifically relevant to this analysis.

The raw dataset was provided in CSV format (Transformed_Housing_Data2.csv) containing 21,609 rows and 31 columns. Key preparation steps had already been applied upstream, including encoding of categorical variables and computation of derived fields like 'Age of House' and 'Years Since Renovation'.

4.2 Data Structure

The dataset is structured as a flat table with each row representing one property transaction. Columns fall into four broad categories:

- Physical attributes: Flat Area, Lot Area, Floors, Bedrooms, Bathrooms, Grade
- Temporal attributes: Age of House (in Years), Years Since Renovation
- Condition & renovation flags: Ever_Renovated_Yes, Condition_of_the_House_* (one-hot encoded)
- Geographic attributes: Latitude, Longitude, Zipcode_Group (one-hot encoded, 9 groups)

4.3 Connecting Data to Tableau

To connect the dataset to Tableau Desktop:

- Step 1: Open Tableau Desktop → Connect → Text File
- Step 2: Navigate to and select Transformed_Housing_Data2.csv
- Step 3: Tableau automatically detects column types; verify numeric fields (Sale_Price, Sqft values) are set to 'Number (decimal)'
- Step 4: Verify binary columns (Ever_Renovated_Yes, Waterfront_View_Yes) are treated as dimensions
- Step 5: Save the connection and proceed to the worksheet

No joins or blends were needed as all required data resided in a single CSV file.

5. Data Preparation & Cleaning

5.1 Preparation Steps

Before building any visualizations, the following preparation activities were completed:

5.1.1 Data Type Verification

All 31 columns were reviewed in Tableau's data source pane to confirm correct data types:

- Numeric fields (Sale_Price, Sqft measurements, Lat/Long) confirmed as 'Number (decimal)'
- Integer fields (Bedrooms, Floors, Grade) confirmed as 'Number (whole)'
- Binary/flag fields kept as Dimensions for filtering
- No date columns were present; age-related fields are numeric

5.1.2 Missing Value Check

A scan of the dataset revealed no significant null values across the 21,609 records. The pre-transformation pipeline had handled missing data upstream. Key observations:

- Basement Area = 0 for properties without a basement (valid, not missing)
- Years Since Renovation = 0 for properties never renovated (Ever_Renovated_Yes = 0)
- All Sale_Price values are positive and within a plausible range ($75K – $1.13M)

5.1.3 Calculated Fields in Tableau

The following calculated fields were created in Tableau to support the visualizations:

Calculated Field	Formula / Description
Sale Price (bin)	Created using Tableau bins on Sale_Price with bin size of 50,000 to group prices into 250K, 300K, 350K… buckets
Renovation Status	IF [Ever_Renovated_Yes] = 1 THEN 'Renovated' ELSE 'Not Renovated' END
Total Sales by Price Bin	SUM([Sale_Price]) aggregated by Sale Price bin and filtered by renovation recency
Count of Records	COUNT([Sale_Price]) — used for the data overview KPI card
Avg. Sale Price	AVG([Sale_Price]) — used for the average price KPI display

5.1.4 Data Filters Applied

Filters were applied at the worksheet and dashboard level to ensure focused analysis:

- Price range filter: $75,000 – $1,129,575 (full dataset range)
- Renovation status filter: interactive filter on dashboard for Renovated vs. Not Renovated
- Age of House filter: used in Scenario 4 to show ages 4, 12, 13, 14, 15 years
- Zipcode Group filter: available for regional drill-down

5.2 Challenges Encountered

- One-hot encoded columns (9 zipcode groups, 4 condition groups) required re-labeling in Tableau for readable legends

- The bin size for Sale_Price required experimentation — final bin size of $50,000 provided the clearest distribution histogram
- Aggregation mode for the grouped bar chart (Scenario 4) needed to be set to SUM rather than COUNT to show meaningful feature counts per age group

6. Data Visualizations

Four key visualization scenarios were designed and built in Tableau to address the project objectives. Each scenario targets a specific analytical question relevant to ABC Company's business needs.

Scenario 1: Overall Data Overview

Type: Summary / KPI Text Table

Purpose

This visualization presents a high-level summary of the full dataset, displaying three critical KPI metrics that establish the scale and nature of the housing data. It serves as the first point of reference for any stakeholder reviewing the dashboard.

Metrics Displayed

Metric	Value	Significance
Count of Records	**21,609**	Dataset size; ensures statistical reliability
Average Sale Price	**$511,619**	Pricing benchmark for ABC Company
Total Basement Area	**38,643,798 sqft**	Scale of the physical housing inventory

Tableau Steps

- Create a new worksheet; drag Sale_Price to the Text shelf and change aggregation to AVG
- Create a second text mark for COUNT([Sale_Price])
- Add SUM([Area of the House from Basement (in Sqft)]) as a third KPI
- Format labels with custom number formatting: currency for price, comma-separated for counts

Insight

With 21,609 records and an average price just above half a million dollars, the dataset is both large enough for statistical analysis and representative of a mid-to-high range housing market.

Scenario 2: Total Sales by Years Since Renovation

Type: Histogram (Bar Chart with Sale Price Bins)

Purpose

This histogram visualizes the total sale value distributed across price bands, with bars color-coded by sale price bins (250K, 300K, 350K … 1100K+). It reveals how renovation recency correlates with total sales volume and price concentration.

Key Observations from the Chart

- The $1.1M+ price bin dominates with $1,309,177,425 in total sales — the tallest bar by far
- The $400K bin is the second highest at $261,450,000 followed by $300K at $231,815,000
- Sales drop off at the $550K–$650K range, suggesting a mid-high price gap
- Recently renovated homes appear predominantly in the higher price bins

Tableau Build Steps

- Right-click Sale_Price in the data pane → Create → Bins; set bin size to 50,000
- Drag 'Sale Price (bin)' to Columns; drag SUM(Sale_Price) to Rows
- Drag 'Sale Price (bin)' to Color on the Marks shelf (automatic color gradient)
- Add SUM(Sale_Price) to Label; format as currency with comma separators
- Set X-axis labels to K notation (250K, 300K…) for readability

Insight

The dominance of the $1.1M+ bin suggests that the highest-value properties (likely renovated and in premium zip codes) generate disproportionate total revenue. Mid-range properties ($300K–$400K) form the bulk of transaction volume by count.

Scenario 3: Distribution of House Age by Renovation Status

Type: Pie Chart

Purpose

This pie chart shows how houses are distributed across different age groups and whether they have been renovated. It provides insight into the age composition of the housing inventory and the prevalence of renovation activity across age cohorts.

Key Observations

- House ages range from 50 to 118 years in the sampled segments
- The distribution is relatively even across age groups (each segment ≈ 8–12% of total)
- Segments labeled 50, 63, 68, 71, 75, 76, 88, 94, 96, and 118 represent distinct age cohorts
- Only 4.2% of all properties (914 out of 21,609) have been renovated, indicating renovation is rare

Tableau Build Steps

- Change the chart type to 'Pie' in the Marks card
- Drag 'Age of House (in Years)' to Color and to Label
- Drag COUNT([Sale_Price]) or SUM(Sale_Price) to Size/Angle
- Format slice labels to show age values with percentage
- Use a diverging or categorical color palette for clear segment differentiation

Insight

The broad age distribution (50–118 years) indicates a mature housing inventory. The low renovation rate (4.2%) suggests significant untapped value for property owners — renovating an older home could substantially boost its market price.

Scenario 4: House Age Distribution by Number of Bathrooms, Bedrooms & Floors

Type: Grouped Bar Chart

Purpose

This grouped bar chart compares three feature counts — number of bathrooms, bedrooms, and floors — across different house age groups (4, 12, 13, 14, and 15 years). It helps identify how modern versus slightly older homes differ in their feature profiles.

Key Observations

Age (Years)	Bedrooms (Avg)	Bathrooms (Avg)	Floors (Avg)
4	~2,100	~1,540	~1,180
12	~1,570	~1,220	~930
13	~1,540	~1,180	~910
14	~1,520	~1,150	~850
15	~1,500	~1,110	~825

- Bedrooms consistently score the highest across all age groups, followed by bathrooms, then floors
- Newer homes (4 years old) have the highest counts across all three features
- Feature counts decrease gradually as house age increases from 4 to 15 years

Tableau Build Steps

- Drag 'Age of House (in Years)' to Columns (filter to show only ages 4, 12, 13, 14, 15)
- Create a combined measure using 'Measure Names / Measure Values' to plot Bedrooms, Bathrooms, and Floors side by side
- Drag 'Measure Names' to Color to create the grouped bars
- Set aggregation to SUM for each measure
- Use the color legend to assign: yellow = Bedrooms, orange = Bathrooms, dark red = Floors

Insight

The consistent decline in feature counts with age suggests that older homes (even within the 4–15 year window) tend to be slightly smaller or less feature-rich. This is consistent with a trend of larger, more feature-rich builds in recent years. Bedroom count remains the dominant feature, reinforcing buyer demand for multi-bedroom properties.

7. Dashboard Overview

7.1 Dashboard Design

All four visualization scenarios were assembled into a single interactive Tableau Dashboard (Dashboard2). The dashboard is designed for executive stakeholders who need a quick but comprehensive view of housing market dynamics.

Design principles applied:

- Single-page layout with all four visualizations visible without scrolling
- Interactive filters at the top (renovation status, price range, age group)
- Consistent color palette: dark bars for premium segments, light bars for mid-range
- Clear chart titles and axis labels for self-explanatory readability

7.2 Tableau Public Link

The dashboard has been published to Tableau Public and is accessible at:

> https://public.tableau.com/app/profile/jay.daftardar/viz/Book2_17729773041440/Dashboard2

7.3 Web Integration (Flask)

The Tableau dashboard is embedded in a Flask web application using the Tableau JavaScript Embed API. The web application consists of three pages:

Page	Route	Description
Home	/	Landing page with project overview, key metrics, and scenario summaries
Dashboard	/dashboard	Embedded live Tableau dashboard + visual breakdown of all 4 scenarios
Team	/team	Team member profiles, project badge, scope and skills

The Tableau embed uses the following JavaScript snippet in the dashboard template:

```
var vizElement = divElement.getElementsByTagName('object')[0];
var scriptElement = document.createElement('script');
scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js';
```

8. Key Findings & Insights

8.1 Renovation Impact on Sale Price

The histogram (Scenario 2) reveals a clear positive correlation between renovation investment and sale price:

- Houses priced above $1.1M collectively generated $1,309,177,425 in total sales — the highest of any price bin
- Properties recently renovated tend to cluster in the higher price bins ($500K+)
- However, only 4.2% of properties in the dataset have ever been renovated — indicating a significant market opportunity

> **Business Implication:** ABC Company should consider prioritizing renovation financing products or advisory services for the 95.8% of non-renovated properties, as renovation demonstrably increases sale value.

8.2 Housing Inventory Age Profile

The pie chart (Scenario 3) shows that the housing inventory spans a wide age range (50–118 years), with each age cohort representing roughly equal proportions of the total. This means:

- No single age group dominates the market — inventory is well distributed
- Very old homes (100+ years) still constitute a meaningful portion of active listings
- Renovated properties are spread across all age groups but represent a small minority in each

8.3 Feature Trends by House Age

The grouped bar chart (Scenario 4) shows consistent patterns across homes aged 4–15 years:

- Bedrooms are always the highest-count feature, followed by bathrooms, then floors
- Feature counts are highest for the newest homes (4 years), declining steadily with age
- This likely reflects increasing living space expectations in recent construction

> **Business Implication:** Marketing materials should emphasize bedroom count as the primary feature driver for newer properties, while positioning renovated older homes on value-for-money and character.

8.4 Price Distribution Insights

- The $250K–$450K price range accounts for the bulk of transaction volume by count (mass market)
- The $1.1M+ range drives outsized total revenue despite fewer transactions
- A notable dip in activity in the $550K–$650K range may indicate a buyer reluctance zone or lack of supply in this segment

9. Conclusion

This project successfully achieved all stated objectives. Using Tableau as the primary visualization tool, the team analyzed 21,609 housing transactions to produce four scenario-specific visualizations, integrated into a single interactive dashboard published on Tableau Public.

The analysis uncovered several key insights for ABC Company:

- Renovation has a measurable positive impact on sale price, yet the vast majority of properties remain unrenovated — a strategic opportunity
- The housing inventory is well-distributed across age groups, with no dominant cohort
- Bedroom count is the leading structural feature across all age groups, and newer homes show higher feature density
- The $300K–$400K price range dominates transaction volume, but the $1.1M+ tier drives total revenue

The deliverables produced include:

- Four Tableau worksheets (Scenario 1 through 4) with appropriate chart types and calculated fields
- One integrated Tableau Dashboard (Dashboard2) published to Tableau Public
- A Flask web application with three pages — Home, Dashboard, and Team — embedding the live dashboard
- This project report documenting the full analytics pipeline from data to insight

9.1 Lessons Learned

- Data preparation in Tableau requires careful management of aggregation types; using SUM vs. AVG vs. COUNT produces dramatically different visual results
- One-hot encoded columns in the raw data needed to be re-interpreted as readable dimensions in Tableau
- Dashboard layout design is an iterative process — the first layout draft was revised twice to improve information flow
- Embedding Tableau in Flask via the JavaScript API requires the dashboard URL to be shared/published publicly

9.2 Future Work

- Incorporate geographic maps using Latitude/Longitude data to visualize price hotspots by location
- Apply predictive modeling (e.g., linear regression) to forecast sale prices based on feature inputs
- Add user authentication to the Flask application for personalized dashboard access
- Extend the dataset with new records to track market trends over time